When Will a Borrower Default?

Survival Analysis in Credit Risk

Luke Munnelly

Should these be the same outcome?

- One borrower defaults after three months.

Hypothetical loans.

Should these be the same outcome?

- One borrower defaults after three months.
- Another defaults after two years.

Hypothetical loans.

Should these be the same outcome?

- One borrower defaults after three months.
- Another defaults after two years.

Both defaulted. The timing is different.

Hypothetical loans.

Principal still outstanding

Illustrative: €12,000 loan; €400 principal repaid each month.

After 3 payments

After 24 payments

Assume default just after payment; ignore interest, fees and recoveries.

Principal still outstanding

Illustrative: €12,000 loan; €400 principal repaid each month.

After 3 payments	€10,800 outstanding
After 24 payments	€2,400 outstanding

Assume default just after payment; ignore interest, fees and recoveries.

Principal still outstanding

Illustrative: €12,000 loan; €400 principal repaid each month.

After 3 payments	€10,800 outstanding
After 24 payments	€2,400 outstanding

Earlier default can occur before much principal is repaid.

Assume default just after payment; ignore interest, fees and recoveries.

What do six months tell us?

Observed for six months; no default so far

What do six months tell us?

Observed for six months; no default so far

$$T_D > 6 \text{ months}$$

Contributes to risk sets while observed.

A binary label needs a horizon

- A classifier can predict default within 12 months.

Training and evaluation labels need appropriate follow-up.

A binary label needs a horizon

- A classifier can predict default within 12 months.
- Six months without default does not establish a 12-month label.

Training and evaluation labels need appropriate follow-up.

Which 12-month labels are known?

Optional explanation

Observed history	Default by month 12?
Default at month 3	
Default at month 24	
No default through month 6	

Hypothetical loans; no earlier competing exit in these histories.

Which 12-month labels are known?

Optional explanation

Observed history	Default by month 12?
Default at month 3	Yes
Default at month 24	No
No default through month 6	

Hypothetical loans; no earlier competing exit in these histories.

Which 12-month labels are known?

Optional explanation

Observed history	Default by month 12?
Default at month 3	Yes
Default at month 24	No
No default through month 6	Unknown

Hypothetical loans; no earlier competing exit in these histories.

Why survival analysis?

- Use when events occur.

Machine-learning methods can also model survival outcomes.

Why survival analysis?

- Use when events occur.
- Accommodate right-censored follow-up under suitable assumptions.

Machine-learning methods can also model survival outcomes.

Why survival analysis?

- Use when events occur.
- Accommodate right-censored follow-up under suitable assumptions.
- Here: Cox for interpretable relative default hazards.

Machine-learning methods can also model survival outcomes.

Choose the event and the clock

Origination $\longrightarrow$ first recorded default

Elapsed time since the loan was issued

Follow-up can end in different ways

ILLUSTRATIVE · Hypothetical loans; months since origination



C: right censoring — active when observation ends.

D: closure — a competing event ending default exposure.

The survival function

First use a hypothetical default-only setting.

$$S(t) = P(T_D > t)$$

The survival function

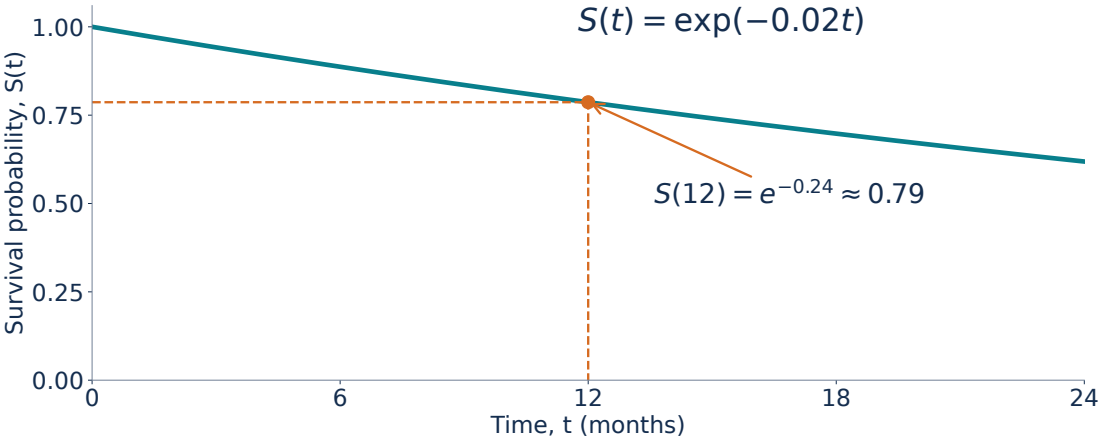
First use a hypothetical default-only setting.

$$S(t) = P(T_D > t)$$

The probability of remaining default-free beyond time t.

Survival: the probability of remaining default-free

HYPOTHETICAL · Default-only setting; not a Bondora estimate



At 12 months, about 79% remain default-free in this hypothetical setting.

How rapidly are defaults occurring now?

Among loans still exposed now:

how rapidly do defaults occur next?

How rapidly are defaults occurring now?

Among loans still exposed now:

how rapidly do defaults occur next?

Hazard: an instantaneous rate, not a probability.

From a short interval to a hazard rate

Hypothetical default-only setting; t in months.

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T_D < t + \Delta t \mid T_D \geq t)}{\Delta t}$$

From a short interval to a hazard rate

Hypothetical default-only setting; t in months.

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \le T_D < t + \Delta t \mid T_D \ge t)}{\Delta t}$$

$$P(t \le T_D < t + \Delta t \mid T_D \ge t) \;\approx\; h(t)\,\Delta t$$

Rate $\times$ a short interval $\approx$ a conditional probability.

From hazard to survival

Optional explanation Default-only setting.

$$H(t) = \int_0^t h(u)\,du$$

From hazard to survival

Optional explanation Default-only setting.

$$H(t) = \int_0^t h(u)\,du$$

$$S(t) = \exp\{-H(t)\}$$

With competing closure, default hazard alone does not determine actual default probabilities.

From timing to borrower characteristics

We have described event timing.

From timing to borrower characteristics

We have described event timing.

How does the default rate differ between borrowers?

The Cox model

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{multiplier}}$$

For credit: a cause-specific default rate among loans still exposed.

The Cox model

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{multiplier}}$$

Baseline: the time pattern for reference covariates.

For credit: a cause-specific default rate among loans still exposed.

The Cox model

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{multiplier}}$$

Baseline: the time pattern for reference covariates.

Multiplier: how the covariates scale that rate.

For credit: a cause-specific default rate among loans still exposed.

Compare otherwise identical borrowers

Hypothetical new vs existing customer; same baseline and other covariates.

$$\mathrm{HR} = \frac{h(t \mid \text{new})}{h(t \mid \text{existing})} = \exp(\beta_{\text{new}})$$

A conditional rate ratio, not a probability ratio or causal effect.

Compare otherwise identical borrowers

Hypothetical new vs existing customer; same baseline and other covariates.

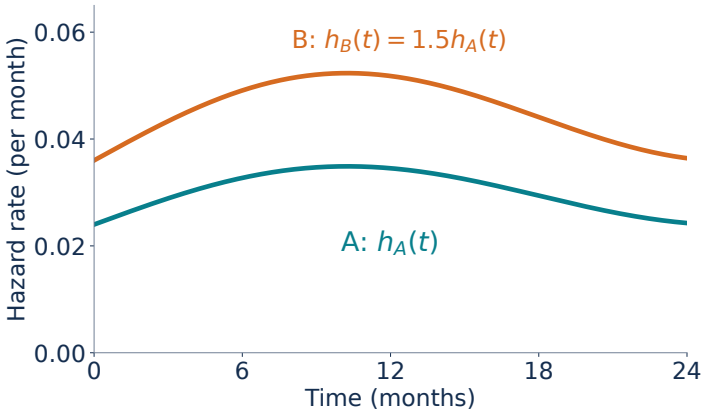
$$\mathrm{HR} = \frac{h(t \mid \text{new})}{h(t \mid \text{existing})} = \exp(\beta_{\text{new}})$$

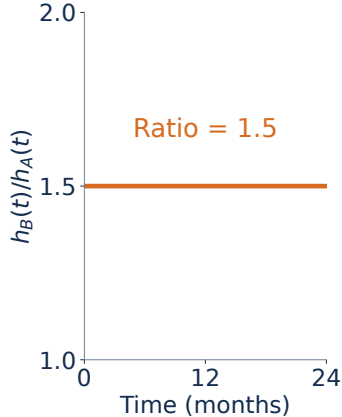
If $\mathrm{HR} = 1.5$: a 50% higher rate at each t, under PH.

A conditional rate ratio, not a probability ratio or causal effect.

Proportional hazards: changing rates, constant ratio

HYPOTHETICAL · A hazard is an instantaneous rate, not a probability





The baseline hazard can vary over time; proportionality concerns the ratio.

How does Cox learn relative effects?

Illustrative: three borrowers at risk just before one default.

A	B	C
Existing customer	Existing customer	New customer
Relative risk 1	Relative risk 1	Relative risk e^{β}

Which borrower accounts for this default?

Cox regression compares borrowers in the risk set

ILLUSTRATIVE · At risk immediately before a default; all other covariates equal

A	B	C
Existing customer	Existing customer	New customer
Relative risk: 1	Relative risk: 1	Relative risk: e^{β}

$$P(C \text{ defaults} \mid \text{one default in risk set}) = \frac{e^{\beta}}{2 + e^{\beta}}$$

If $\beta = \log(2)$: $\quad \frac{2}{1+1+2} = \frac{1}{2}$

Conditional on one default occurring now; not a probability over a time horizon.

The audited Bondora cohort

Estonia · Issued 2013–2015

Follow-up: at most 730 days from origination

Reporting cutoff: 20 July 2021; all loans have 730 days of potential follow-up.

The audited Bondora cohort

Estonia · Issued 2013–2015

Follow-up: at most 730 days from origination

9,844	6,924	2,508
Loans	Borrowers	Recorded defaults

Reporting cutoff: 20 July 2021; all loans have 730 days of potential follow-up.

Follow-up uses audited endpoints

- Default: first recorded `DefaultDate`.

Closure is not guaranteed full cash repayment. The source default threshold is unverified.

Follow-up uses audited endpoints

- Default: first recorded `DefaultDate`.
- Closure proxy: `ContractEndDate` only when snapshot `Status=Repaid`.

Closure is not guaranteed full cash repayment. The source default threshold is unverified.

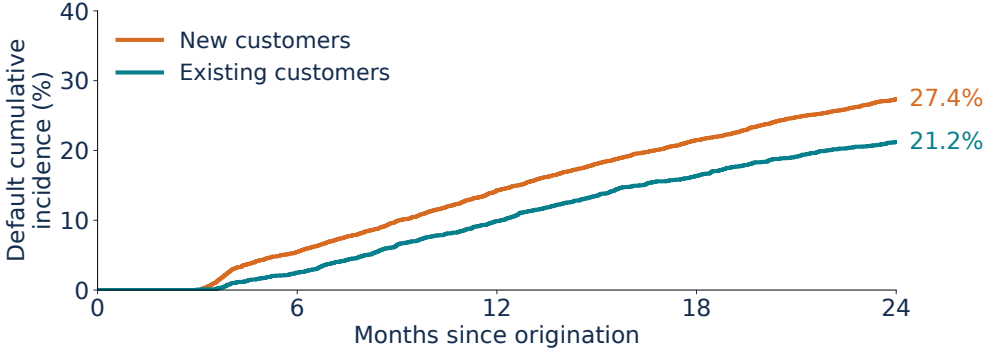
Follow-up uses audited endpoints

- Default: first recorded `DefaultDate`.
- Closure proxy: `ContractEndDate` only when snapshot `Status=Repaid`.
- Stop at the first exit or 730 days.

Closure is not guaranteed full cash repayment. The source default threshold is unverified.

When do recorded defaults accumulate?

UNADJUSTED / DESCRIPTIVE · Closure proxy treated as a competing event



Loans at risk just before the indicated month

New	6,839	5,971	4,855	3,842	3,081
Existing	3,005	2,683	2,253	1,851	1,534

Aalen–Johansen · No uncertainty bands · Month = 730/24 days · Not a causal comparison

Five predictors and their contrasts

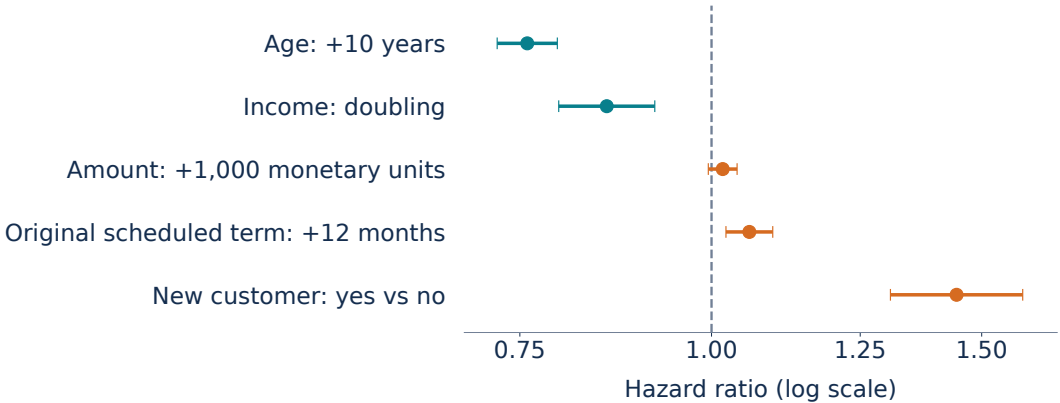
Age	+10 years
Income	Doubling
Amount	+1,000 recorded monetary units
Original scheduled term	+12 months
New-customer status	Yes vs no

Year-stratified baseline; borrower-clustered intervals.

Monthly original-schedule proxy; source currency/income period unverified.

Adjusted associations with the default rate

ADJUSTED · Audited cause-specific Cox model; origination-year baseline strata



Conditional associations · Borrower-clustered 95% CIs; approximate with tied dates.

PH concerns: age/new customers. Source monetary currency/income period unverified.

New versus existing customer

$$\text{HR} = 1.445$$

95% CI: 1.308–1.595

New versus existing customer

$$\mathrm{HR} = 1.445$$

95% CI: 1.308–1.595

About 44% higher conditional default rate among loans still exposed.

Association, not a default-probability ratio or causal effect.

PH concern: this single HR is a model summary.

Where interpretation is limited

- PH concerns: age and new-customer status.

Where interpretation is limited

- PH concerns: age and new-customer status.
- Nonlinearity; approximate tied-date intervals.

Where interpretation is limited

- PH concerns: age and new-customer status.
- Nonlinearity; approximate tied-date intervals.
- Recorded endpoints and measurement proxies.

When will a borrower default?

- No certain individual default date.

When will a borrower default?

- No certain individual default date.
- By 24 months: 27.4% new vs 21.2% existing.

Timing proportions are unadjusted/descriptive under the closure proxy.

When will a borrower default?

- No certain individual default date.
- By 24 months: 27.4% new vs 21.2% existing.
- Adjusted new-customer HR: 1.445.

Timing proportions are unadjusted/descriptive under the closure proxy.

HR is a conditional association, not a probability ratio or causal effect.

Questions

Backup: partial likelihood

$$L(\beta) = \prod_{j:\,\text{default}} \frac{\exp(\beta^{\mathsf{T}} x_{i_j})}{\sum_{k \in \mathcal{R}_{y_j}(t_j)} \exp(\beta^{\mathsf{T}} x_k)}$$

$\mathcal{R}_{y_j}(t_j)$: loans still exposed immediately before t_j, within the defaulting loan's origination-year stratum.

Schematic formula for distinct event times. The audited fit uses Efron handling of ties.

Backup: corrected diagnostics

	PH rank p	PH log p	Nonlinear p
Age	0.00641	0.00768	0.00358
Log income	0.933	0.621	0.185
Amount	0.236	0.118	0.805
Original term	0.777	0.997	0.0438
New customer	0.00888	0.000417	—

PH: approximate independence-based Schoenfeld tests, not cluster-adjusted. Rank uses elapsed event times. Nonlinearity: existing spline cluster-Wald checks; exploratory.

One loan per borrower: 6,924 loans, 1,842 defaults. Broadly similar coefficient directions; sample composition changes.